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Identification apparatus and image forming apparatus

Abstract

An identification apparatus includes a guide member defining a guide path, a scanning portion that scans the sheet inserted in the guide path, a first detection portion disposed upstream of the scanning portion in an insertion direction of the sheet in the guide path that detects presence/absence of the sheet, and a second detection portion disposed downstream of the scanning portion in the insertion direction in the guide path and detects presence/absence of the sheet. The identification apparatus is configured to start measurement of a surface property of the sheet by the scanning portion in response to a change in a detection result of one of the first detection portion or the second detection portion, and complete the measurement in response to a change in a detection result whichever of the first detection portion or the second detection portion not used in determining the start of the measurement.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/161,727, filed on Jan. 30, 2023, which claims priority from Japanese Patent Application No. 2022-018465, filed on Feb. 9, 2022, which are hereby incorporated by reference herein in their entireties.

BACKGROUND

Field
(1) The present disclosure relates to an identification apparatus that measures a surface property of

a sheet and to an image forming apparatus.

Description of the Related Art

(2) In recent years, various types of sheets have been used as sheets serving as recording materials in image forming apparatuses such as copiers, printers, facsimile machines, and multifunctional apparatuses having functions of these based on demand from users. For example, in an image forming apparatus of an electrophotographic system, optimum conditions in each step such as charging, exposure, development, transfer, or fixation vary depending on the type of the sheet. Therefore, it is desired that the type of the sheet on which image formation is to be performed is appropriately set in advance. However, for example, there is a case where the user does not know the type of the sheet, and there is a possibility that the type of the sheet cannot be correctly set in the image forming apparatus. Japanese Patent Laid-Open No. 2016-109425 proposes identifying the type of the sheet by measuring surface properties of the sheet by an optical sensor when a user inserts the sheet.

(3) In recent years, the variety of the types of sheets being used has become more diverse due to the demand for diversity in on-demand printing, and it is desired that the measurement precision is improved also for the identification apparatus that identifies the type of the sheet as described above. However, in an identification apparatus that performs measurement of a sheet inserted by a user, for example, there is a case where the user fails to correctly insert the sheet. In such a case, for example, if the sheet does not move and the sheet is scanned by an optical sensor at the same position, the measurement data thereof is insufficient, and there is a possibility that the measurement precision becomes poor.

SUMMARY OF THE INVENTION

(4) According to an aspect of the present disclosure, an identification apparatus includes a guide member configured to define a guide path through which a sheet inserted therein is guided, a scanning portion configured to scan the sheet inserted in the guide path, a first detection portion disposed upstream of the scanning portion in an insertion direction of the sheet in the guide path and configured to detect presence/absence of the sheet, and a second detection portion disposed downstream of the scanning portion in the insertion direction in the guide path and configured to detect presence/absence of the sheet. The identification apparatus is configured to start measurement of a surface property of the sheet by the scanning portion in response to a change in a detection result of one of the first detection portion or the second detection portion, and complete the measurement in response to a change in a detection result of whichever of the first detection portion or the second detection portion not used in determining the start of the measurement.

(5) Further features will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an image forming apparatus including a sheet type identification apparatus according to a present embodiment.

(2) FIG. 2 is a block diagram illustrating a control system according to the present embodiment.

(3) FIG. 3A is a diagram illustrating a state in which a sheet is separated from a heating roller of a fixing portion.

(4) FIG. 3B is a diagram illustrating a state in which a sheet is wound around the heating roller of the fixing portion.

(5) FIG. 4A is a section view of the sheet type identification apparatus in a state before the sheet according to the present embodiment is inserted.

(6) FIG. 4B is a schematic diagram illustrating the sheet type identification apparatus in the state

before the sheet according to the present embodiment is inserted as viewed from above.

(7) FIG. 5A is a section view of the sheet type identification apparatus in a state in which the sheet according to the present embodiment is inserted.

(8) FIG. 5B is a schematic diagram illustrating the sheet type identification apparatus in the state in which the sheet according to the present embodiment is inserted as viewed from above.

(9) FIG. 6 is a schematic diagram illustrating the sheet type identification apparatus in the state in which the sheet according to the present embodiment is inserted as viewed in an insertion direction.

(10) FIG. 7 is a flowchart illustrating measurement processing of the sheet type identification apparatus according to the present embodiment.

(11) FIG. 8 is a diagram illustrating a starting screen of sheet setting according to the present embodiment.

(12) FIG. 9 is a diagram illustrating a relationship between an ultrasonic wave transmission coefficient and grammage.

(13) FIG. 10 is a diagram illustrating a measurement error screen according to the present embodiment.

(14) FIG. 11 is a flowchart illustrating sheet type identification processing according to the present embodiment.

(15) FIG. 12 is a diagram illustrating a relationship between an adjacent pixel difference accumulation value, a brightness, and a sheet type.

(16) FIG. 13 is a diagram illustrating a sheet selection screen for sheet setting according to the present embodiment.

(17) FIG. 14 is a diagram illustrating a brand determination screen in the sheet selection screen for sheet setting according to the present embodiment.

(18) FIG. 15 is a diagram illustrating sheet type data according to the present embodiment.

(19) FIG. 16 is a diagram illustrating sheet reference data in which a relationship between a sheet brand and a reference value of a measurement item according to the present embodiment is recorded.

DESCRIPTION OF THE EMBODIMENTS

(20) An embodiment of the present disclosure will be described in detail below with reference to the corresponding drawings. FIG. 1 illustrates a schematic configuration of the image forming apparatus according to an exemplary embodiment. More specifically, an image forming apparatus **201** including a sheet type identification apparatus **100** as an identification apparatus according to the present disclosure will be described with reference to FIG. 1.

(21) Schematic Configuration of Image Forming Apparatus

(22) As illustrated in FIG. 1, reference number **201** denotes an image forming apparatus, reference number **201A** denotes an image forming apparatus body, and reference number **201B** denotes an image forming portion that forms an image on a sheet. Reference number **202** denotes an image reading apparatus that is approximately horizontally disposed above the image forming apparatus body **201A**, and a discharge space V for discharging sheets is provided between the image reading apparatus **202** and the image forming apparatus body **201A**. An operation portion **502** serving as an operation display portion that is constituted by a touch panel or the like and capable of receiving an inputting operation and displaying a screen, and a sheet type identification apparatus **100** that will be described in detail below are provided above the image forming apparatus body **201A**, that is, in an upper portion of the image forming apparatus **201**.

(23) The image forming portion **201B** serving as image forming portion is a four-drum full-color system. The image forming portion **201B** includes a laser scanner **210**, and four process cartridges **211Y**, **211M**, **211C**, and **211K** that respectively form toner images of four colors of yellow (Y), magenta (M), cyan (C), and black (K). Each of the process cartridges **211Y**, **211M**, **211C**, and **211K** include a photosensitive drum **212**, a charging unit **213** serving as a charging means, and a developing unit **214** serving as a developing portion. In addition, the image forming portion **201B**

includes an intermediate transfer unit **201C** and a fixing portion **201E** that are disposed above the process cartridges **211Y**, **211M**, **211C**, and **211K**. Reference number **215** denotes a toner cartridge for supplying toner to the developing unit **214**.

(24) The intermediate transfer unit **201C** includes an intermediate transfer belt **216** looped over a driving roller **216a** and a tension roller **216b**. Primary transfer rollers **219** that abut the intermediate transfer belt **216** at positions respectively opposing the photosensitive drums **212** are provided on the inner peripheral side of the intermediate transfer belt **216**. The intermediate transfer belt **216** is rotated in a direction indicated by an arrow by the driving roller **216a** driven by driving portion (not illustrated).

(25) The primary transfer rollers **219** sequentially transfer toner images of respective colors having a negative polarity on the photosensitive drums **212** onto the intermediate transfer belt **216** to be superimposed on each other. A secondary transfer roller **217** that transfers the color image formed on the intermediate transfer belt **216** onto the sheet S is provided at a position opposing the driving roller **216a** of the intermediate transfer unit **201C**. A secondary transfer portion **201D** is formed between the intermediate transfer belt **216** and the secondary transfer roller **217**. A fixing portion **201E** including a pressurizing roller **220a** and a heating roller **220b** is disposed above the secondary transfer roller **217**. In addition, a first discharge roller pair **225a**, a second discharge roller pair **225b**, and a duplex inversion portion **201F** are disposed on the upper-left side of the fixing portion **201E**. The duplex inversion portion **201F** includes a reverse conveyance roller pair **222** capable of rotating in a normal direction and a reverse direction, a re-conveyance path R for conveying a sheet on one surface of which an image has been formed to the image forming portion **201B** again, and the like.

(26) A sheet feeding unit **230** that delivers a set sheet S to the image forming portion **201B** is provided in a lower portion of the image forming apparatus body **201A**. The sheet feeding unit **230** includes a feed cassette **1** serving as a sheet supporting portion that accommodates and supports the sheet S, and a sheet feeding portion **5** that feeds the sheet S accommodated in the feed cassette **1**. The sheet feeding portion **5** includes a pickup roller **2**, and a feed roller **3** and a retard roller **4** serving as a separation portion for separating sheets S that are simultaneously delivered by the pickup roller **2** from each other.

(27) A manual feeding portion **235** that delivers the set sheet S to the image forming portion **201B** is provided on the right side surface of the image forming apparatus body **201A** in FIG. **1**. The manual feeding portion **235** includes a manual feed tray **6** serving as a sheet supporting portion that supports the sheet S, and includes a sheet feeding means and a separation portion similarly to the sheet feeding unit **230**.

(28) Next, an image forming operation of the image forming apparatus **201** will be described. When image information of a document is read by the image reading apparatus **202**, the image information is subjected to image processing, and is then converted into an electric signal and transmitted to the laser scanner **210** of the image forming portion **201B**. In the image forming portion **201B**, the surfaces of the photosensitive drums **212** are each uniformly charged to a predetermined polarity and a predetermined potential by the charging unit **213** and are sequentially exposed by laser light. As a result, yellow, magenta, cyan, and black electrostatic latent images are sequentially formed on the photosensitive drums **212** of respective process cartridges **211**.

(29) These electrostatic latent images are developed and visualized with the toners of respective colors, and the toner images of respective colors on the respective photosensitive drums **212** are sequentially transferred onto the intermediate transfer belt **216** by primary transfer bias applied to the primary transfer rollers **219** to be superimposed on each other. As a result, a toner image is formed on the intermediate transfer belt **216**.

(30) The sheet S fed by the feed roller **3** of the sheet feeding unit **230** is conveyed to a registration roller pair **240** constituted by a driving roller and a driven roller (not illustrated). Driving of the registration roller pair **240** is stopped, and the leading end of the sheet S abuts the registration roller

pair **240**. As a result, the leading end of the sheet S is caused to follow the registration roller pair **240**. The feed roller **3** continues conveying the sheet S, which results in forming of a warpage (loop) in the sheet S, and the registration roller pair **240** is driven when a predetermined loop amount is reached. As a result, the skew of the sheet S is corrected by the registration roller pair **240**, and the sheet S whose skew has been corrected is conveyed to the secondary transfer portion **201D** by the registration roller pair **240**. Sequentially, in the secondary transfer portion **201D**, the toner image is collectively transferred onto the sheet S by a secondary transfer bias applied to the secondary transfer roller **217**. Then, the sheet S onto which the toner image has been transferred is conveyed to the fixing portion **201E**, toner of respective colors is melted and mixed by the heat and pressure from the fixing portion **201E**, and is fixed to the sheet S as a color image.

(31) The sheet S to which the image has been fixed is discharged to the discharge space V by a first discharge roller pair **225a** and a second discharge roller pair **225b** provided downstream of the fixing portion **201E**, and is supported on a supporting portion **223** formed on the bottom surface of the discharge space V. In the case of forming images on both surfaces of the sheet S, after the image is fixed, the sheet S is conveyed to the re-conveyance path R by the reverse conveyance roller pair **222**, and is conveyed to the image forming portion **201B** again.

(32) Control System of Image Forming Apparatus and Sheet Type Identification Apparatus

(33) FIG. 2 is a block diagram illustrating a control system according to the present embodiment. More specifically, a control system of the image forming apparatus **201** and the sheet type identification apparatus **100** will be described with reference to FIG. 2.

(34) As illustrated in FIG. 2, the image forming apparatus **201** includes a main controller **400** serving as a controller that controls the operation of the image forming apparatus **201**, the operation of the sheet type identification apparatus **100** electrically connected to the image forming apparatus **201**, and the like. The image forming apparatus **201** includes a memory **401** constituted by a random access memory (RAM) or the like, a sheet type database **402** constituted by a read-only memory (ROM) or the like, and the operation portion **502** that are each electrically connected to the main controller **400**.

(35) The sheet type identification apparatus **100** includes a controller **160** connected to the main controller **400** of the image forming apparatus **201** described above. The sheet type identification apparatus **100** also includes an upstream sheet sensor **104**, a downstream sheet sensor **105**, an optical sensor **130**, an ultrasonic wave sensor **135**, and a sheet thickness sensor **140** that are connected to the controller **160** and that are described below.

(36) Necessity of Setting Sheet Type in Image Forming Apparatus

(37) FIG. 3A illustrates a state where a sheet is separated from a heating roller of a fixing portion. FIG. 3B illustrates a state where the sheet is wound around the heating roller of the fixing portion. More specifically, the necessity of setting the sheet type when performing an image forming operation in the image forming apparatus **201** will be described with reference to FIGS. 3A and 3B.

(38) Conditions of the image forming operation, that is, the conveyance speed and temperature for the fixation, the transfer voltage of the secondary transfer, and the like in printing vary depending on the type of the sheet on which an image is to be formed. Examples of the type of the sheet include physical properties such as the grammage, rigidity, surface property, material, and the like. Therefore, the type of the sheet to be used needs to be set in advance for the image forming operation.

(39) Some sheets have restrictions to be set in the feed cassette **1**, the manual feed tray **6**, and the like that serve as a sheet feeding port. For example, some cardboards with high rigidity can only be fed from the manual feed tray **6** whose conveyance path has a low curvature. Coated paper sheets that have a smooth surface and a tendency of adhering to each other need to be fed one by one from the manual feed tray **6**. Paper sheets formed from pulp typically have different flexural rigidity in the longitudinal direction and the lateral direction due to unevenness in the orientation direction of the pulp fiber caused by the manufacturing method. The orientation direction is referred to as a

paper grain direction. Therefore, in some cases, there is a recommended direction in the longitudinal or lateral direction for setting the sheet in the feed cassette **1** or the manual feed tray **6** such that the flexural rigidity for bending in the conveyance path is low. In some cases, for a one-side coated paper sheet that is a plain paper sheet whose one surface is coated, which surface of the sheet should face upward when being set in the feed cassette **1** or the manual feed tray **6** is specified for performing printing on the coated surface.

(40) Some types of sheets cannot be used in the image forming apparatus **201**. For example, for cardboards with excessively high rigidity, the conveyance of the sheet can stop due to resistance that the sheet receives when being conveyed through a bent conveyance path. For thin paper sheets having excessively low rigidity, the sheet is strongly affected by an adhesive force between molten toner and the heating roller **220b** when passing through the fixing portion **201E**. In the case of a sheet such as a plain paper sheet, as illustrated in FIG. 3A, after the sheet is heated and pressurized by the heating roller **220b** and the pressurizing roller **220a**, the leading end of the sheet is separated from the heating roller **220b** and the pressurizing roller **220a** due to the rigidity thereof. Therefore, the sheet can be separated from the heating roller **220b** by a separation plate **220c**. In the case of a thin paper sheet having excessively low rigidity, as illustrated in FIG. 3B, after the sheet is heated and pressurized by the heating roller **220b** and the pressurizing roller **220a**, the leading end of the sheet does not separate from the heating roller **220b**, and thus the sheet cannot be separated from the heating roller **220b** by the separation plate **220c**. Therefore, a malfunction in which the sheet is wound around the heating roller **220b** can occur. In addition, in the case of a synthetic sheet formed from synthetic resin instead of pulp, there is a possibility that the synthetic sheet melts when subjected to high heat in the fixing portion **201E** and contaminates the heating roller **220b**.

(41) As described above, to perform image formation appropriately without causing a failure in image formation by the image forming apparatus **201**, it is important to identify the type of the sheet to be used before the image formation is started. Then, in the image forming apparatus **201**, the conditions of the image formation need to be appropriately set, or declining acceptance of the sheet is needed depending on the type of the sheet.

(42) Configuration of Sheet Type Identification Apparatus

(43) The configuration of the sheet type identification apparatus **100** will be described with reference to FIGS. 4A to 6. FIG. 4A is a section view of a sheet type identification apparatus in a state before the sheet according to the present embodiment is inserted. FIG. 4B is a schematic diagram illustrating the sheet type identification apparatus in the state before the sheet according to the present embodiment is inserted as viewed from above. FIG. 5A is a section view of the sheet type identification apparatus in a state in which the sheet according to the present embodiment is inserted. FIG. 5B is a schematic diagram illustrating the sheet type identification apparatus in the state in which the sheet according to the present embodiment is inserted as viewed from above. FIG. 6 is a schematic diagram illustrating the sheet type identification apparatus in the state before the sheet according to the present embodiment is inserted as viewed in the insertion direction.

(44) The sheet type identification apparatus **100** is provided together with the operation portion **502** on the image forming apparatus body **201A** as illustrated in FIG. 1. When the user inserts the sheet S in the guide path **101**, the sheet type identification apparatus **100** measures and obtains the physical properties of the sheet S. The measurement portion for the physical properties is constituted by a mechanical property measurement portion **102** and a surface property measurement portion **103**, and the physical properties of the sheet S such as the grammage, the surface property, the sheet thickness, and the like can be obtained when the sheet S passes the mechanical property measurement portion **102** and the surface property measurement portion **103**.

(45) As illustrated in FIG. 4A, the sheet type identification apparatus **100** includes an upper block **109** serving as a guide member or a first guide member and a lower block **110** serving as a guide member or a second guide member. The upper block **109** includes an upper guide surface **109a**. The lower block **110** includes a lower guide surface **110a** and a pressing member **106** that will be

described below includes a pressing surface **106a**. The lower guide surface **110a** and the pressing surface **106a** are disposed to oppose the upper guide surface **109a**, and thus the guide path **101** for guiding the inserted sheet S is defined. The upper block **109** includes an abutment portion **108** that the sheet S inserted in the guide path **101** abuts. The pressing member **106** is urged and pressed against the upper guide surface **109a** or an optical sensor **130**, described below, by a spring **107** serving as an urging member and provided in the lower block **110** in a contracted state.

(46) As illustrated in FIGS. **4A**, **4B**, and **6**, the mechanical property measurement portion **102** includes an ultrasonic wave sensor **135** serving as a grammage detection portion that measures the grammage of the inserted sheet S as illustrated in FIG. **2**. The mechanical property measurement portion **102** includes a sheet thickness sensor **140** serving as a thickness detection portion that measures the thickness of the inserted sheet S as illustrated in FIG. **2**. The surface property measurement portion **103** includes an optical sensor **130** that measures the surface properties such as the surface roughness, gloss, and the like as illustrated in FIG. **2**.

(47) As illustrated in FIGS. **4A** and **4B**, the sheet type identification apparatus **100** includes an upstream sheet sensor **104** serving as a first detection portion or a movement detection portion, and a downstream sheet sensor **105** serving as a second detection portion or a movement detection portion. The upstream sheet sensor **104** outputs a different signal for a detection result depending on the presence or absence of the sheet S in a detection position X1 upstream of the mechanical property measurement portion **102** and the surface property measurement portion **103** in the insertion direction X of the sheet S. That is, the upstream sheet sensor **104** detects the presence or absence of the sheet S. The downstream sheet sensor **105** outputs a different signal for a detection result depending on the presence or absence of the sheet S in a detection position X2 downstream of the mechanical property measurement portion **102** and the surface property measurement portion **103** in the insertion direction X of the sheet S. That is, the downstream sheet sensor **105** detects the presence or absence of the sheet S. The upstream detection position X1 is located near the entrance of the guide path **101**, and the downstream detection position X2 is located near the abutment portion **108** of the guide path **101**. That is, the upstream sheet sensor **104** detects insertion of the sheet S into the sheet type identification apparatus **100**, and the downstream sheet sensor **105** detects abutment of the sheet S on the abutment portion **108**. The upstream sheet sensor **104** and the downstream sheet sensor **105** can detect the movement of the sheet S as a result of having different detection timings for the sheet S, and thus, constitute a movement detection portion.

(48) Grammage Measurement

(49) The ultrasonic wave sensor **135** of the mechanical property measurement portion **102** includes an ultrasonic wave transmitter **136** disposed in the lower block **110** and an ultrasonic wave receptor **137** disposed in the upper block **109** such that the guide path **101** is interposed therebetween. The ultrasonic wave sensor **135** communicates an ultrasonic wave via the sheet S inserted in the guide path **101**, and detects the grammage of the sheet S. The grammage is the mass of the sheet S per unit area, and can be expressed by [g/m.sup.2].

(50) The ultrasonic wave transmitter **136** and the ultrasonic wave receptor **137** are each constituted by a piezoelectric element that is an element that converts mechanical displacement and an electric signal into each other and an electrode terminal. In the ultrasonic wave transmitter **136**, when a pulse voltage of a predetermined frequency is input to the electrode terminal, the piezoelectric element oscillates to generate an ultrasonic wave, and the ultrasonic wave is transmitted through the air. When the ultrasonic wave reaches the sheet S, the sheet S is vibrated by the ultrasonic wave. As described above, the ultrasonic wave generated by the ultrasonic wave transmitter **136** is transmitted to the ultrasonic wave receptor **137** via the sheet S. The piezoelectric element of the ultrasonic wave receptor **137** generates, in the electrode terminal, an output voltage corresponding to the amplitude of the ultrasonic wave received in accordance with the grammage. Since the output voltage of the case where the ultrasonic wave is transmitted via the sheet S is attenuated as compared with a case where the sheet S is not present between the ultrasonic wave transmitter **136**

and the ultrasonic wave receptor **137**, the transmittance is calculated from this ratio of attenuation. Since the transmittance of the ultrasonic wave changes based on the thickness of the sheet S, the grammage of the sheet S can be estimated by using an ultrasonic wave transmittance coefficient-sheet grammage conversion formula. This is the operation principle of detecting the grammage of the sheet S by communicating an ultrasonic wave using piezoelectric elements.

(51) Measurement of Sheet Thickness

(52) The sheet thickness sensor **140** of the mechanical property measurement portion **102** is a lever type sensor, and detects a value corresponding to the thickness of the sheet S by an encoder (not illustrated). The lever pivots based on the thickness of the sheet S, and the encoder detects the value corresponding to the thickness of the sheet S by rotating based on the amount of pivot in an interlocked manner with the lever. The sheet thickness sensor **140** measures the sheet thickness where the sheet S is detected by the downstream sheet sensor **105**, and thus the abutment of the sheet S on the abutment portion **108** is detected. The sheet thickness sensor **140** is disposed to be positioned slightly upstream of the detection position X2 of the downstream sheet sensor **105** in the insertion direction X.

(53) Measurement of Surface Property

(54) The optical sensor **130** of the surface property measurement portion **103** includes a light-emitting diode (LED) **131** serving as a light source and a contact image sensor (CIS) **132** serving as a scanning portion that are disposed on the upper block **109** side as illustrated in FIGS. 4A, 4B, and 6. The optical sensor **130** measures the surface property of the sheet S by irradiating the surface of the sheet S with light from the LED **131** and receiving the light reflected based on the unevenness of the surface of the sheet S by the CIS **132** via a condenser lens (not illustrated).

(55) In the measurement of the surface property of the optical sensor **130**, since the surface of the sheet S needs to be held at the optical focusing position, the orientation of the sheet S is stabilized by the pressing member **106** urged by the spring **107** as described above. As a result, the variation in the position and orientation of the sheet S at the time of measuring the surface of the sheet S by the optical sensor **130** is reduced, and the measurement of the surface property can be stably performed. The spring **107** is set such that the inserted sheet S is set to be pressed against the optical sensor **130** by a force of approximately 100 gf.

(56) The scanning direction at the scanning position of the CIS **132** is positioned orthogonal to the insertion direction X of the sheet S. That is, in the CIS **132**, dots of image sensors are arranged in the scanning direction, and imaging can be performed at a resolution corresponding to the pitch of the dots. In the present embodiment, the image sensors are arranged at a pitch of 300 dpi. The CIS **132** can only read an image of one line in one scanning, and this is insufficient for identifying the surface property of the sheet S. In this case, the unevenness of the sheet S detected by the CIS **132** is too local, and if the surface property is determined from an image of only one line in the surface of the sheet S, deviation of the output result at each measurement position is large, and the data of one line is not appropriate for determination of the surface property. To address this, the overall tendency of the surface property of the sheet S can be determined by using a result obtained by performing one-line scanning a plurality of times in the surface of the sheet S and averaging the data of respective lines. To achieve this, the measurement can be performed while the sheet S is relatively moving with respect to the optical sensor **130**. That is, dynamic measurement can be performed.

(57) To support this dynamic measurement, in the present embodiment, the upstream sheet sensor **104** and the downstream sheet sensor **105** that detect the sheet S are respectively provided upstream and downstream of the optical sensor **130** of the surface property measurement portion **103** in the insertion direction X in the guide path **101**. The relationship between the detection by the sensors and the measurement of the sheet S will be described in the description of the measurement processing by the sheet type identification apparatus **100**.

(58) The local detection of the unevenness by the CIS **132** enables determination of the

characteristics of the surface property for a sheet with deep recesses, such as an embossed paper sheet. While the dynamic measurement is needed to determine the overall tendency also in this case, the identification of the embossed paper sheet is made possible by using the CIS **132**, and thus the identification of the sheet can be performed more flexibly.

(59) Measurement Processing by Sheet Type Identification Apparatus

(60) The measurement processing executed by the controller **160** of the sheet type identification apparatus **100** will be described with reference to FIGS. **7**, **8**, **9**, and **10**. FIG. **7** is a flowchart illustrating measurement processing by the sheet type identification apparatus **100** according to the present embodiment. FIG. **8** is a diagram illustrating a starting screen of the sheet setting according to the present embodiment. FIG. **9** is a diagram illustrating a relationship between the transmission coefficient of the ultrasonic wave and the grammage. FIG. **10** is a diagram illustrating a measurement error screen according to the present embodiment.

(61) In the case where a user uses a sheet, the main controller **400** of the image forming apparatus **201** displays a sheet selection screen illustrated in FIG. **8** on the operation portion **502**. If a user selects “Automatic Sheet Identification”, the main controller **400** causes the sheet type identification apparatus **100** to begin executing the processing illustrated in the flowchart of FIG. **7**.

(62) In step **S1**, the sheet type identification apparatus **100** transitions to an automatic identification mode. Next, in step **S2**, the main controller **400** commands the controller **160** to start measurement. In step **S3**, the controller **160** of the sheet type identification apparatus **100** performs initialization processing of the mechanical property measurement portion **102** and the surface property measurement portion **103** to prepare for insertion of the sheet **S**. Next, in step **S4**, the user inserts the sheet **S** in the guide path **101** of the sheet type identification apparatus **100**. In step **S5** a determination is made whether the upstream sheet sensor **104** has detected the presence of a sheet. If no sheet is detected, (No), the processing returns to step **S1**, and the process of steps **S1-S4** are repeated until the upstream sheet sensor **104** detects insertion of the sheet **S**. If a sheet is detected (when the upstream sheet sensor **104** detects insertion of the sheet **S**) (Yes), a timer **t** starts counting time from $t=0$.

(63) After the elapse of a first predetermined time ($t=t_1$) from the detection of sheet **S** by the upstream sheet sensor **104** as a trigger ($t=0$), in step **S6**, the surface of the sheet **S** is measured by scanning the surface of the sheet **S** a plurality of times by the optical sensor **130** for each predetermined cycle based on a predetermined sequence. The measurement result is processed by an ASIC of the controller **160**, and the processed measurement result is accumulated in a memory in the controller **160** as surface property data. In the present embodiment, as the surface property data, information of an accumulation value of difference between adjacent pixels and information of brightness are generated and accumulated.

(64) After the elapse of a second predetermined time ($t=t_2$) from the detection of sheet **S** by the upstream sheet sensor **104** as a trigger ($t=0$), in step **S7**, the transmittance of the ultrasonic wave through the sheet **S** is measured by the ultrasonic wave sensor **135** based on a predetermined sequence. The controller **160** converts the transmittance into grammage information, and the grammage information is generated and accumulated in the memory in the controller **160** similarly to the surface property data. To convert the transmittance of ultrasonic wave into grammage, the grammage is determined by using an ultrasonic wave transmittance coefficient-grammage conversion formula indicated by a broken line in FIG. **9**.

(65) In step **S8**, the controller **160** determines whether a timeout time ($t=t_{out}$) has elapsed as a set time of the timer **t** from the detection of the sheet **S** by the upstream sheet sensor **104** as a trigger ($t=0$). Before the timeout time of the timer **t** elapses, that is, when a result of step **S8** is “Yes”, the processing proceeds to step **S10**. In the case where the downstream sheet sensor **105** has detected the sheet **S** before the timeout time of the timer **t** elapses, that is, in the case where the result of step **S10** is “Yes”, the sheet thickness of the sheet **S** is measured by the sheet thickness sensor **140** in step **S11**. The sheet thickness information is generated and accumulated in a memory in the

controller **160**.

(66) As a result of the downstream sheet sensor **105** detecting the sheet S before the timeout time, the controller **160** determines that the sheet S has moved while being measured by the optical sensor **130**. That is, as a result of the scanning being performed at a plurality of different positions to which the sheet S has moved instead of the same position, data with high precision can be obtained, and it is determined that accurate data is obtained.

(67) The controller **160** determines that all measurement has been successfully completed, and notifies the main controller **400** of the image forming apparatus **201** that the measurement has been successfully finished. Then, in step **S12**, the controller **160** transmits the grammage, surface property, and information of the sheet thickness serving as measurement data that has been generated and accumulated to the main controller **400**. The controller **160** then finishes the measurement processing, and the main controller **400** transitions to sheet type identification processing for identifying the sheet type.

(68) Returning to step **S8**, in the case where the timeout time of the timer t has elapsed before the downstream sheet sensor **105** detects the sheet S, that is, in the case where the result of step **S8** is No, processing of stopping the measurement as an error is performed in step **S9**. That is, the controller **160** determines that accurate data of the surface property of the sheet is not obtained, discards all data accumulated in the memory in the controller **160**, and transmits error information to the main controller **400** of the image forming apparatus **201**. Then, the main controller **400** displays a screen prompting re-insertion of the sheet in the sheet type identification apparatus **100** for re-measurement on the operation portion **502** as illustrated in FIG. **10**. After the user recognizes this instruction for re-insertion, the process returns to step **S1**, and the measurement processing illustrated in FIG. **7** is repeated.

(69) As described above, in the measurement processing according to the present embodiment, the measurement by the optical sensor **130** started in response to detection of the sheet S by the upstream sheet sensor **104**, and the measurement by the optical sensor **130** is completed in response to detection of the sheet S by the downstream sheet sensor **105**. As a result, it can be confirmed that the scanning of the sheet S by the optical sensor **130** has been performed while the sheet S is moving. That is, it can be confirmed that the surface property of the sheet is dynamically performed, and thus the precision of the measurement of the surface property of the sheet can be improved.

(70) In the measurement processing according to the present embodiment, the measurement is stopped as an error in the case where the sheet S is not detected by the downstream sheet sensor **105** after the sheet S is detected by the upstream sheet sensor **104** and before the timeout time elapses. This means that the insertion of the sheet S by the user has stopped halfway, and the sheet S is not inserted to abut the abutment portion **108**. Since there is a possibility that the movement of the sheet S has stopped halfway, there is a possibility that the scanning of the sheet S by the optical sensor **130** is performed at the same position and the measurement is not performed successfully. Therefore, by stopping the measurement as an error, erroneous identification of the type of the sheet S can be suppressed, and by prompting the user to perform measurement again, the precision of the measurement of the surface property of the sheet can be improved.

(71) Sheet Type Identification Processing

(72) Next, sheet type identification processing executed by the main controller **400** of the image forming apparatus **201** will be described with reference to FIGS. **11** to **16**. FIG. **11** is a flowchart illustrating the sheet type identification processing according to the present embodiment. FIG. **12** is a diagram illustrating a relationship between the accumulation value of difference between adjacent pixels, the brightness, and the sheet type. FIG. **13** is a diagram illustrating a sheet selection screen for sheet setting according to the present embodiment. FIG. **14** is a diagram illustrating a brand identification screen in the sheet selection screen for sheet setting. FIG. **15** is a diagram illustrating sheet type data according to the present embodiment. FIG. **16** is a diagram illustrating sheet

reference data in which a relationship between sheet brands and reference values of measurement items according to the present embodiment are recorded.

(73) As described above, in the case where it is determined by the controller **160** that accurate measurement data has been obtained and the measurement data is transmitted, the main controller **400** identifies the type of the sheet by performing sheet type identification processing.

(74) Turning to FIG. **11**, the main controller **400** classifies sheet types in step **S21** based on the measurement data of the surface property. As an index of the surface property, a data table (data matrix) whose vertical axis is indicated by the adjacent pixel difference accumulation value, whose horizontal axis is indicated by the brightness, and in which classification has been performed in advance by measuring the surface properties of characteristic sheets is used as illustrated in FIG. **12**. The adjacent pixel difference accumulation value is a value obtained by accumulating luminance difference between adjacent pixels in the CIS **132**, and is an indicator of the unevenness of the surface of the sheet **S**. The brightness is an accumulation value of the luminance values of the pixels of the CIS **132** that has received light, and represents the brightness of the sheet **S**.

(75) For example, a transparent film formed from resin such as polyethylene terephthalate: PET does not reflect much light from the LED **131**, and thus the measured brightness thereof is low. In the case of a sheet to the surface of which geometrical recesses and protrusions are intentionally provided, such as an embossed paper sheet, the luminance difference between adjacent pixels becomes large due to the recesses and protrusions, and therefore the adjacent pixel difference accumulation value becomes large. In the case of a recycled paper sheet, the grain direction of the sheet is inconsistent, the pulp fibers are shorter as a result of undergoing the recycle process including a plurality of steps, and therefore there is a tendency that the surface property of the sheet is measured as rough. In the case of a coated paper sheet, the unevenness of the surface appears to be low due to the coating layer on the surface, and there is a tendency that the adjacent pixel difference accumulation value becomes small. In this manner, the surface properties of the sheet can be classified based on luminance information of each pixel of the CIS **132**, and thus the sheet can be classified based on sheet classification used in the image forming apparatus **201**. As described above, the main controller **400** determines, based on the received measurement data, which class in the data table the adjacent pixel difference accumulation value and brightness of the sheet correspond to, and classifies the sheet type based on the surface property of the sheet.

(76) In step **S22**, determination is made based on the grammage of the sheet **S** in the received measurement data. That is, the type of the sheet is identified by adding the grammage information to the classification of the sheet made based on the surface property. For example, in the case where a sheet of GF-C081 manufactured by Canon Marketing Japan Inc., the sheet is identified by the sheet type identification apparatus **100**, the sheet is identified in a category called Plain Paper **2**.

(77) In step **S23**, determination is made based on the sheet thickness of the sheet **S** in the received measurement data. That is, by using the sheet thickness information in the measurement data, the estimation precision of the grammage is improved. For example, in the case of a coated paper sheet, the surface of a medium quality paper sheet or the like is coated with a mixture of a white pigment such as clay (kaolin) or calcium carbonate and an adhesive (binder) such as starch as a paint. The coated paper sheet has a higher density than plain paper sheets for printing that include pulp as a main component, and this affects the error in the ultrasonic wave transmittance coefficient-grammage conversion formula. Therefore, the identification precision of the sheet can be improved by using the sheet thickness information together. It is effective to add the information of the sheet thickness to identify the brand of the sheet.

(78) Sheet Selection

(79) Next, the processing in step **S24** of determining the sheet type information of the sheet **S** set in the image forming apparatus **201** in the present embodiment will be described. As illustrated in FIG. **13**, the image forming apparatus **201** stores types of sheet settings that are selectable for each of classes set in advance. For example, the sheet GF-C081 described above is a sheet included in a

sheet class of Plain Paper 2 in terms of nominal grammage, but can be classified into a different sheet class due to the variation in the grammage of the sheet itself. Therefore, a sheet class close to the class identified based on the measurement data, that is, a sheet type different from the identified sheet type is also displayed as a candidate. The sheet class identified based on the measurement by the sheet type identification apparatus **100** this time, that is, the identified sheet type is displayed as recommended. When the user performs an operation to select one of the sheet classes displayed in this manner, that is, when the main controller **400** receives the operation input, the selected sheet class is set as the type of the sheet S to be used for the image formation by the image forming apparatus **201**.

(80) In the case where the user selects “To Brand Identification” on the lower-left side of the screen illustrated in FIG. **13**, the screen transitions to a brand identification screen, illustrated in FIG. **14**, for selecting the brand of the sheet as an example of a sheet selection screen. The user selects the brand of the sheet while checking brands on the brand identification screen illustrated in FIG. **14**, and the selected sheet brand is set as the type of the sheet S to be used for image formation by the image forming apparatus **201**.

(81) As described above, by selecting the sheet from the list of sheet classes or sheet brands, the user can perform the printing with appropriate printing settings by assigning the sheet settings (setting of sheet type) to the feed cassette **1** and the manual feed tray **6**.

(82) For example, in the case where a sheet class or sheet brand of the sheet S intended by the user is not in the sheet selection screen illustrated in FIG. **13** or FIG. **14**, “Perform Automatic Sheet Identification Again” on the lower-right side of the either screen can be selected to transition to the measurement processing illustrated in FIG. **7**.

(83) The sheet type database **402** illustrated in FIG. **2** stores sheet type data in which the physical properties, parameters of optimum image formation conditions, compatibility with the image forming apparatus **201**, usable feeding port (feedable port), and the like for each brand of the sheet are described, as illustrated in FIG. **15**. The sheet type database **402** illustrated in FIG. **2** also stores sheet reference data in which reference values of the grammage, surface properties (adjacent pixel difference accumulation value and brightness), and sheet thickness that can be measured by the sheet type identification apparatus **100** are associated with the brand information, as illustrated in FIG. **16**. The main controller **400** refers to the sheet type data and the sheet reference data, and displays several candidates having high similarity on the operation portion **502**. Thus, the user can easily select the type of the sheet S and perform the sheet setting. Since the sheet class is also determined if the sheet brand is determined, by determining the sheet settings of the image forming apparatus **201** from the sheet type data of FIG. **15** and the sheet reference data of FIG. **16**, the image formation (printing) can be performed at the optimum quality.

(84) Other Embodiments

(85) The above-described sheet type identification apparatus **100** is merely an example and the above-described configuration is not seen to be limiting. In the present embodiment, the sheet type identification apparatus **100** includes the controller **160**, and information of the optical sensor **130**, information of the ultrasonic wave sensor **135**, and information of the sheet thickness sensor **140** are processed and transmitted to the main controller **400**. In another exemplary embodiment, the measurement data obtained by each sensor can be transmitted to the main controller **400** as is, and various processing can be performed by the main controller **400**. While a case where the main controller **400** determines the type of the sheet has been described, the above-described configuration is not seen to be limiting. In another exemplary embodiment, the type of the sheet can be determined by the controller **160** of the sheet type identification apparatus **100**. In this case, providing, in the sheet type identification apparatus **100**, the sheet type database **402** which is included in the image forming apparatus **201** in the above example can be considered.

(86) The present embodiment described using detection of the sheet S by the upstream sheet sensor **104** as a trigger for the start of measurement. In another exemplary embodiment, the detection of

the sheet S by the downstream sheet sensor **105** can be used as the trigger for the start of measurement. In this case, the measurement is performed when pulling the sheet S out from the sheet type identification apparatus **100**. That is, the measurement of the surface property of the sheet by the CIS **132** is started in response to change in the detection result of one of the upstream sheet sensor **104** or the downstream sheet sensor **105**. Then, the measurement can be completed in response to change in the other of the upstream sheet sensor **104** or the downstream sheet sensor **105**. In addition, an activated state in which the measurement of the sheet in a standby state by the optical sensor **130** is possible can be taken by using the detection of the sheet S by the upstream sheet sensor **104** as a trigger.

(87) The above-described description of the present embodiment was provided assuming that the measurement is treated as an error if a timeout time has elapsed from the detection of the sheet S by the upstream sheet sensor **104**. In another exemplary embodiment, the measurement can be treated as an error in the case where, for example, the upstream sheet sensor **104** detects the presence of the sheet S and then the upstream sheet sensor **104** detects the absence of the sheet S without the downstream sheet sensor **105** detecting the presence of the sheet S.

(88) The present embodiment provides an example in which various control parameters of the image forming operation by the image forming apparatus **201** are determined by selecting a sheet class or sheet brand based on the characteristics detected by the sheet type identification apparatus **100**. In another exemplary embodiment, various control parameters of the image forming operation can be directly set from the feature values of the sheet measured by the sheet type identification apparatus **100**.

(89) While the present embodiment has been described with the sheet type database **402** and the main controller **400** provided in the image forming apparatus **201**, in another exemplary embodiment, one or both can be provided external to the image forming apparatus **201**. For example, the sheet type database **402** and the main controller **400** can be cloud based, where the latest sheet type setting information and the latest identification algorithm can be selected if the image forming apparatus **201** is connected to the cloud.

(90) While the present embodiment has been described with the sheet type identification apparatus **100** disposed in the image forming apparatus **201**, in another exemplary embodiment, the sheet type identification apparatus **100** is disposed external to the image forming apparatus **201**.

(91) The present embodiment describes a case where the lower guide surface **110a** and the pressing surface **106a** of the pressing member **106** are provided to oppose the upper guide surface **109a**. In another exemplary embodiment, the lower guide surface **110a** is removed and the pressing surface **106a** of the pressing member **106** is opposed to and pressed against the entirety of the upper guide surface **109a**. In another exemplary embodiment, the lower guide surface **110a** of the lower block **110** is fixed, the pressing member is disposed in the upper block **109**, and the pressing surface of the pressing member is pressed against the lower guide surface **110a**. In this case, the optical sensor **130** is preferably disposed in the lower block **110**. That is, the optical sensor **130** is disposed in either the upper block **109** or the lower block **110**, and a surface is formed on one side in the thickness direction of the guide path **101**. In still yet another exemplary embodiment, a surface on the other side in the thickness direction of the guide path **101** is formed in the other of the upper block **109** or the lower block **110**, and an urging member that urges the surface on the other side toward the surface on the one side can be employed.

(92) The present embodiment describes a case where the ultrasonic wave transmitter **136** is disposed in the lower block **110** and the ultrasonic wave receptor **137** is disposed in the upper block **109**. In another exemplary embodiment, the ultrasonic wave transmitter **136** is disposed in the upper block **109** and the ultrasonic wave receptor **137** is disposed in the lower block **110**. That is, a configuration where the ultrasonic wave transmitter **136** is disposed on one side of the guide path **101** in the thickness direction of the sheet S and the ultrasonic wave receptor **137** is disposed on the other side of the guide path **101** in the thickness direction of the sheet can be employed.

(93) The present embodiment describes a case where the upstream sheet sensor **104** and the downstream sheet sensor **105** are used for detecting the movement of the sheet S during measurement by the optical sensor **130**. In another exemplary embodiment, any element can be used as a movement detection portion that detects movement of the sheet S in the scanning position of the CIS **132**. For example, a configuration in which the movement of the sheet S is detected by using a roller that is rotated by following the sheet S and an encoder that detects the rotation of the roller can be employed. In another exemplary embodiment, configuration in which the movement of the sheet S is read by an image reading apparatus or a line sensor capable of reading in the insertion direction X can be employed. In another exemplary embodiment, a configuration where, among the above-described upstream sheet sensor **104** and the downstream sheet sensor **105**, the upstream sheet sensor **104** is not provided and only the downstream sheet sensor **105** is provided, and the upstream sheet sensor **104** is substituted by the CIS **132** (optical sensor **130**) can be employed. That is, in this case, the movement of the inserted sheet in the scanning position of the scanning portion is detected by the downstream sheet sensor **105** serving as a movement detection portion detecting the sheet S within a set time from the detection of the sheet S by the CIS **132**. In addition, the measurement is stopped as an error in the case where the set time elapses before the sheet S is detected by the downstream sheet sensor **105** after the CIS **132** has detected the sheet S.

(94) Other Embodiments

(95) Embodiment(s) can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(96) While exemplary embodiments have been described, these embodiments are not seen to be limiting. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. An identification apparatus with an opening that is provided on one end in an insertion direction and an abutment portion against which a sheet abuts and that is provided on another end in the insertion direction, the identification apparatus comprising: a reading sensor configured to read a surface of the sheet; a first detection portion configured to detect presence/absence of the sheet; and a second detection portion configured to detect presence/absence of the sheet, wherein the first detection portion, the reading sensor, and the second detection portion are aligned in this order in the insertion direction.
2. The identification apparatus according to claim 1, wherein, when reading by the reading sensor

is completed, a result of the reading by the reading sensor is output.

3. The identification apparatus according to claim 1, wherein, in a case where a set time has elapsed after reading by the reading sensor is started and before a detection result of the first detection portion of the second detection portion not used in determining the start of the reading changes, an error condition is determined and the reading is stopped.
4. The identification apparatus according to claim 3, wherein when the reading by the reading sensor is stopped, information of the error is output.
5. The identification apparatus according to claim 1, wherein the reading sensor starts reading in response to detection of the sheet by the first detection portion and stops the reading in response to detection of the sheet by the second detection portion.
6. The identification apparatus according to claim 1, further comprising: a grammage detection portion configured to detect a grammage of the sheet, wherein detection by the grammage detection portion is started in response to detection of the sheet by the first detection portion.
7. The identification apparatus according to claim 6, wherein the grammage detection portion includes an ultrasonic wave transmitter disposed on one side of the guide path in a thickness direction of the sheet and an ultrasonic wave receptor disposed on a second side of the guide path in the thickness direction of the sheet.
8. The identification apparatus according to claim 1, further comprising: a thickness detection portion configured to detect a thickness of the sheet inserted in the guide path, wherein detection by the thickness detection portion is started in response to detection of the sheet by the second detection portion.
9. The identification apparatus according to claim 1, wherein the sensor starts reading in response to the first detection portion changing from a state of detecting the sheet to a state of not detecting the sheet, and stops the reading in response to the second detection portion changing from a state of detecting the sheet to a state of not detecting the sheet.
10. The identification apparatus according to claim 1, wherein the reading sensor includes a light source, and a contact image sensor configured to scan a surface of the sheet irradiated by the light source.
11. The identification apparatus according to claim 1, further comprising: a first guide member on which the reading sensor is disposed and configured to constitute one surface of the guide path in a thickness direction of the sheet, a second guide member disposed to oppose the first guide member and configured to constitute a second surface of the guide path in the thickness direction, and an urging member configured to urge the second guide member toward the first guide member.
12. An image forming apparatus comprising: the identification apparatus according to claim 1; a sheet supporting portion configured to support the sheet having undergone the reading of the surface of the sheet by the identification apparatus; and an image forming portion configured to form an image on the sheet supported by the sheet supporting portion.
13. The image forming apparatus according to claim 12, further comprising a controller configured to identify a type of the sheet based on information of a result of the reading of the surface of the sheet by the identification apparatus.
14. The image forming apparatus according to claim 13, further comprising: an operation display portion configured to display an image and receive an operation input, wherein the controller controls the operation display portion to display a screen to prompt to perform the reading of the surface of the sheet by the identification apparatus in response to input of error information from the identification apparatus.
15. The image forming apparatus according to claim 14, wherein, in response to identification of the type of the sheet based on the information of the result of the reading of the surface of the sheet, the controller controls the operation display portion to display the identified type of the sheet and a type of the sheet different from the identified type of the sheet as candidates, and receives input of an operation of selecting a type of sheet from the displayed candidates.

16. The image forming apparatus according to claim 12, wherein the identification apparatus is provided in an upper portion of the image forming apparatus.
